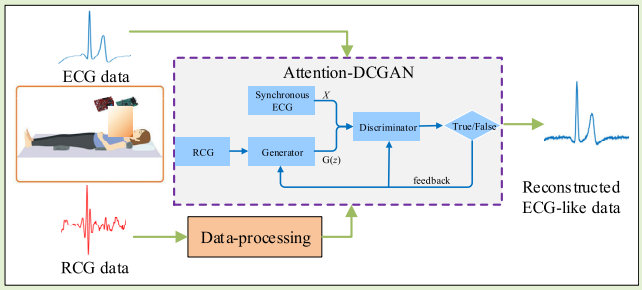
Reconstructing ECG-Like Signals From Millimeter-Wave Radar Echoes: A Generative Adversarial Network Approach

Biao Jin, Hao Wu, Yi Wang, Zhenkai Zhang, Xiangqun Zhang, and Genyuan Du

***Abstract*—Electrode-based electrocardiogram (ECG) measurement devices often lead to user discomfort and raise privacy concerns. In contrast, millimeter-wave radar offers a noninvasive alternative by enabling real-time heartbeat monitoring without direct physical contact. Nevertheless, radar outputs differ significantly from traditional ECG signals, complicating medical diagnosis. To address this challenge, we propose a method for accurately reconstructing ECG-like signals from millimeter-wave radar data using deep convolutional generative adversarial networks (DCGANs) enhanced with an attention mechanism (Attention-DCGAN). Based on the MMECG public dataset, we first apply an improved *K*-means algorithm to cluster heartbeat data from various heart locations. We then train the Attention-DCGAN, consisting of a generator with a four-layer deconvolution module and a one-layer attention module, alongside a discriminator with a four-layer convolution module, to convert radar-derived heartbeat data into ECG-like signals. Experimental results demonstrate that our method achieves higher reconstruction accuracy than traditional methods, with a root-mean-square error of 0.0329 mV and a Pearson correlation coefficient (PCC) of 0.9853.**



***Index Terms*— Attention mechanism, electrocardiogram (ECG), generative adversarial networks, *K*-means, millimeter-wave radar, vital signs monitoring.**

I. Introduction

With the global increase in the aging population, there is a pressing need to meet the societal demands for elderly care, healthcare, and medical assistance [1]. Many older adults with heart disease require continuous monitoring of their heart activity. The electrocardiogram (ECG), a critical biomedical signal, is extensively utilized to describe human cardiac activity and serves as a fundamental tool in diagnosing heart disease [2], [3]. As depicted in Fig. 1, the ECG signal comprises three primary waves: P, QRS, and T. Normally, the *Q*, *R*, and S waves aggregate to form the QRS complex, and the interval between these waves is a vital parameter in ECG analysis. The segment between the S wave and the T wave signifies the transition into slow ventricular diastole; the T wave's smaller amplitude and extended duration indicate potential changes during this phase. The heart rate, another significant parameter, is calculated based on the intervals between consecutive *R* peaks, providing critical insight into cardiac function [4], [5], [6]. Traditionally, ECG signals are collected using bioelectrical techniques, where electrodes on the skin's surface detect cardiac potentials. However, this contact-based method involves adhesive electrodes, potentially infringing on patient privacy, causing discomfort, and reducing compliance [7], [8].

To overcome these issues, noncontact vital signs monitoring technologies have been developed [9], [10]. Millimeter-wave radar, noted for its low power consumption, strong anti-interference capability, and high resolution, has been employed in noncontact monitoring of vital signs, particularly respiration and heart rates [11], [12]. This technology measures heart rate parameters by capturing electromagnetic signals reflected

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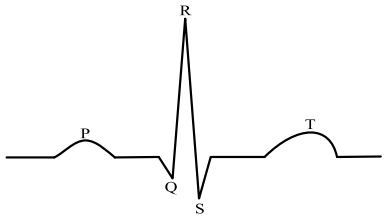


Fig. 1. ECG signal. The P wave signifies the potential shift during atrial contraction. The QRS wave represents the potential change during ventricular contraction. The T wave indicates the potential shift during ventricular diastole.

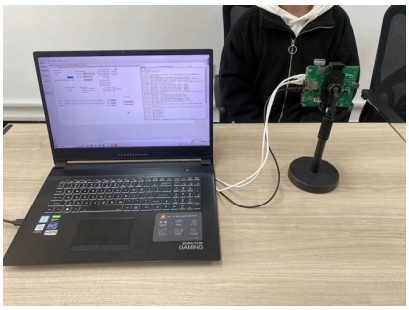


Fig. 2. Millimeter-wave radar for monitoring human heartbeat.

from the body and extracting phase information corresponding to vibrations over time [13], [14]. These vibrations, as illustrated in Fig. 2, are detected by millimeter-wave radar, which transmits electromagnetic waves in the millimeter band to the chest and extracts phase information from subtle heart vibrations at specific instances. Benefits of this approach include eliminating direct body contact and facilitating wide-area monitoring. Nonetheless, environmental factors such as breathing and physical movements can introduce noise, affecting measurement accuracy. Therefore, effective signal processing and filtering techniques are required to ensure reliable results when using millimeter-wave radar for heartbeat information acquisition [15], [16].

A. Related Work

The exploration of contactless sensors for detecting human breathing and heartbeat has garnered significant attention in recent years. For instance, Mercuri et al. [17] employ frequency-modulated continuous wave (FMCW) radar to monitor the heartbeat and respiration of multiple individuals simultaneously. This noncontact method, which eliminates the need for wearable components, employs a pair of antennas (transmitter and receiver) along with sophisticated algorithms for target tracking and filtering out random body movements. Consequently, it demonstrates resilience against moderate limb movements and maintains effective tracking during vigorous activities such as walking and standing up. Similarly, Park et al. [18] examine the capabilities of ultra-wide bandwidth (UWB) radar in measuring respiratory rate (RR), pulse rate (PR), and RR intervals concurrently. Their experimental findings support the UWB radar's ability to monitor respiration, carotid pulse, and heart rhythm simultaneously, indicating its suitability for continuous, noncontact vital signs monitoring in clinical settings.

Furthermore, Li et al. [19] introduce a motion-robust heartbeat sensing technique using joint unsupervised source separation for 4-D imaging radar. By leveraging multiview measurements of asymmetric motion in 4-D radar images, their method integrates physical modeling and data-driven insights. Real-world experiments validate its efficacy across various body motions, such as swaying and random movements, achieving a remarkable 92.2% accuracy in heartbeat signal extraction.

The capability of millimeter-wave radar to capture the minutiae of cardiac micro-movements has been established. However, radar-based cardiographs (RCGs) differ significantly from traditional ECG signals, limiting their utility in medical diagnostics. Current research primarily targets heart rate detection [16], [20], [21], [22], with a limited investigation into the relationship between RCG and ECG signals. A notable study by Chen et al. [23] utilize the radar reflection signal modulated by cardiac activity to scan the heart. They developed a specialized signal processing algorithm to derive cardiac motion signals and compiled MMECG datasets. The proposed deep neural network, structured around a data-driven encoder-decoder framework, has demonstrated a compelling potential for ECG signal reconstruction, achieving a Pearson correlation of 90% with the original ECG and a root mean square error (RMSE) of 0.081 mV. Despite these promising results, there remains an imperative to enhance the reconstruction accuracy further.

B. Our Contribution

In this article, we propose a deep convolutional generative adversarial network (DCGAN) with an attention mechanism to tackle the heart-related domain transformation problem, enabling end-to-end reconstruction mapping from radar ECG input to ECG-like output. Validation on open datasets demonstrates the enhanced accuracy of this approach for applications employing radar technology for ECG measurement. The main contributions of this work are as follows.

1) Based on the open MMECG datasets, we enhance the distinguishability and stability of millimeter-wave radar data by applying an improved K-means clustering algorithm focused on 50 3-D locations of heart signals. This method reduces time-domain fluctuations and improves the accuracy of millimeter-wave radar heartbeat measurements.
2) We develop a DCGAN augmented with an attention mechanism (Attention-DCGAN) to reconstruct ECG-like signals from RCG signals. An attention mechanism layer is integrated following the last deconvolution layer of the generator network, establishing a precise mapping between RCG and ECG signals for accurate ECG-like waveform reconstruction.

The rest of this article is organized as follows. Section II describes the processing flow of millimeter-wave radar measurements of heartbeat signals. Section III presents the architecture of the generative adversarial network (GAN). Section IV verifies the effectiveness of the proposed method through experiments. Finally, Section V concludes this article.

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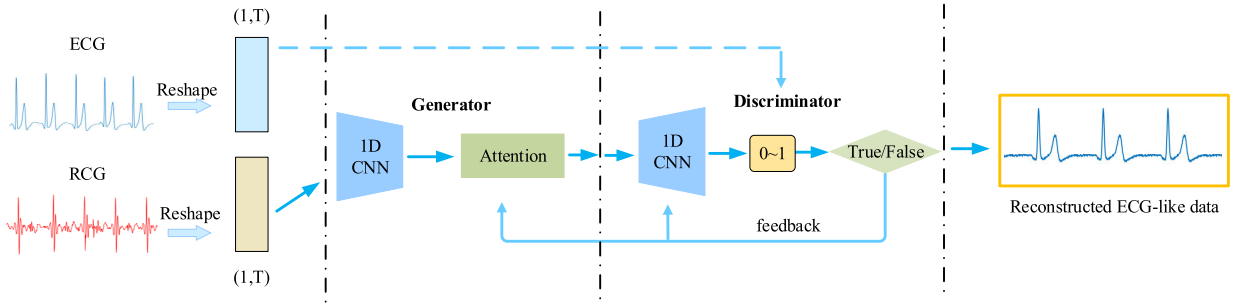


Fig. 3. Basic flow of the proposed method.

II. Signal Model

The FMCW signal emitted by the millimeter-wave radar to the human chest can be expressed as follows:

$$s(t) = \exp\left[j\left(2\pi f_c t + \pi \frac{B}{T} t^2\right)\right] \tag{1}$$

where f_c is the carrier frequency, B is the signal bandwidth, and T is the period of the transmitted signal.

The echo signal of the human chest is the delayed signal of the transmitted signal

$$r(t) = \exp\left[j\left(2\pi f_c (t-\tau) + \pi \frac{B}{T}(t-\tau)^2\right)\right] \tag{2}$$

where the time delay $\tau = 2R/c$ with R being the distance of the human chest relative to the radar and c being the propagation speed of electromagnetic waves in free space. The radar echo signal from the human chest is mixed and filtered, and then the intermediate frequency (IF) signal can be expressed as follows:

$$\begin{aligned} b(t) &= s^*(t) r(t) \approx \exp\left[j\left(4\pi \frac{BR}{cT} t + \frac{4\pi}{\lambda} R\right)\right] \\ &= \exp\left[j(2\pi f_b t + \phi_b)\right] \end{aligned} \tag{3}$$

where $(\cdot)^*$ denotes the conjugate, and λ is the wavelength of the signal. f_b and ϕ_b are the frequency and phase of the IF signal, respectively. They are represented as follows:

$$f_b = \frac{2BR}{cT} \tag{4}$$

and

$$\phi_b = \frac{4\pi R}{\lambda}. \tag{5}$$

We assume that the displacement produced by the heartbeat is $d(t)$, so the actual distance relative to the radar is $R + d(t)$. Thus, (4) and (5) can be rewritten as follows:

$$f_b = \frac{2B(R + d(t))}{cT} \tag{6}$$

and

$$\phi_b = \frac{4\pi (R + d(t))}{\lambda}. \tag{7}$$

To obtain the chest vibration information $d(t)$ from the radar echo, it is crucial to measure the phase variation over time. The phase difference signal between the neighboring pulses in a given time interval is

$$\Delta\phi_b(t) = \frac{4\pi}{\lambda} d(t). \tag{8}$$

As shown in (8), the phase difference reflects the displacement caused by the human heartbeat. Therefore, by measuring the frequency of the phase difference, the frequency of the human heartbeat can be obtained.

III. Reconstruction From RCG to ECG

In fact, RCG signals also contain valuable information about cardiac physiological activities. Therefore, we can obtain more detailed information about cardiac activities similar to ECG signals, which will be an essential basis for analyzing and researching chronic heart diseases. Based on the millimeter-wave radar public dataset MMECG, we first cluster the RCG signals at 50 3-D locations to obtain RCG signals synchronized with the ECG signals. Next, we employ a DCGAN to reconstruct the ECG-like signals, aiming to approximate the morphological features of the ECG signals. Fig. 3 illustrates the specific process.

A. Improved K-Means Clustering

The extraction of cardiac motion signals using millimeter-wave radar heavily relies on the phase extraction of orthogonal I/Q signals. However, the nonlinear operation may result in the loss of signal power information. To improve the data quality, we propose a spatial filter based on K-means clustering to spatially merge and filter the heartbeat data acquired by millimeter-wave radar. The filtering process considers the position, reflected energy, and time domain signal correlation. We employ the K-means clustering algorithm to cluster the data with predefined N nonoverlapping clusters for merging signals related to heart motion. During the filtering process, we merge the signals within each cluster toward its center of mass [24]. The clustering function is defined as follows:

$$J = \sum_{i=1}^{M} \sum_{j=1}^{N} \left(w_{i,j} \rho_s \left\| s_i - g_j \right\|^2 + w'_{i,j} \rho_l \left\| l_i - l_{\mu_j} \right\|^2\right) \tag{9}$$

where M and N are dot number and cluster number, respectively. s_i is the phase difference signal of the ith dot, g_j is the mass center of the jth cluster, l_i is the 3-D position of

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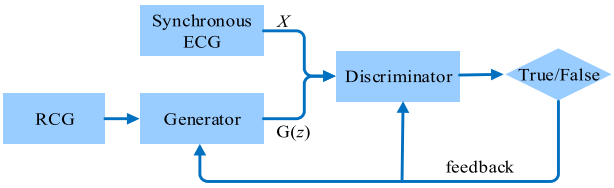


Fig. 4. GAN model.

the ith dot, and l_{μ_j} is the 3-D position of the mass center of the jth cluster. $w_{i,j}$ and $w'_{i,j}$ are the phase difference weight and position weight of the ith dot in the jth cluster, respectively. We employ the expectation-maximization (EM) method to solve the weights. ρ_s and ρ_l are the equilibrium ratios of the phase difference and the position, respectively. The positions of the clustered center of mass g_j and center of mass l_{u_j} are updated by weighted normalization of the motion signal s_i and the signal power p_i at the corresponding position l_i, respectively. The expressions of g_j and l_{u_j} are given as follows:

$$g_j = \frac{\sum_{i=1}^{M} w_{i,j}\, p_i s_i}{\sum_{i=1}^{M} w_{i,j}\, p_i} \tag{10}$$

and

$$l_{\mu_j} = \frac{\sum_{i=1}^{M} w'_{i,j}\, p_i l_i}{\sum_{i=1}^{M} w'_{i,j}\, p_i} \tag{11}$$

where p_i is the signal power of the ith dot.

B. Reconstruction Networks Based on Attention-DCGAN

We propose a GAN architecture to reconstruct ECG signals from radar-based Cardiograph signals. Our approach leverages a deep convolutional GAN integrated with an attention mechanism, termed Attention-DCGAN, to effectively map RCG signals to ECG-like signals. The incorporation of the attention mechanism into the network aims to enhance its focus on critical features, utilizing deep convolutional networks for accurate ECG signal reconstruction and synthesis [25], [26].

1) Reconstruction Principle: The GAN comprises the generator (G) and the discriminator (D), as shown in Fig. 4. The generator takes the RCG signals as input. It generates ECG-like signals through learning. In contrast, the discriminator serves as a binary classifier that distinguishes between input data being an actual ECG signal or generated by the generator. The GAN can be viewed as a game-theoretic model involving the generator and discriminator networks. The generator aims to deceive the discriminator by enhancing its ability to generate data resembling the ECG signal.

Meanwhile, the discriminator improves its discriminative capabilities by learning to differentiate between the ECG signal and the generated data accurately. This dynamic interaction between the generator and discriminator allows both networks to enhance their respective skills progressively. As a result, the generator becomes more proficient in generating data that closely approximates the ECG signal. In contrast, the discriminator becomes more accurate at distinguishing between the real ECG signal data and generated data.

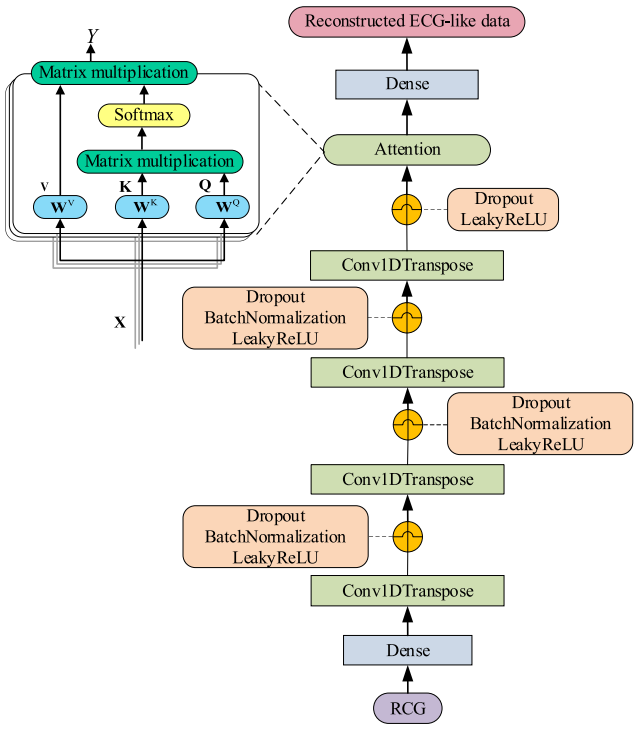


Fig. 5. Structure of the generator.

We introduce the attention mechanism into the GAN and construct Attention-DCGAN to enhance the network's attention to important information. In this way, Attention-DCGAN can generate more accurate ECG-like signals and achieve reconstruction mapping from RCG to ECG signals.

2) Network Structure of Generator: Fig. 5 depicts the network structure of the proposed generator. The RCG signal, as the input, is passed through a dense layer. This layer connects four Conv1DTranspose layers [27] to perform the anti-convolution operation. Each anti-convolutional layer has a distinct number of neurons and parameter settings to gradually expand the data's dimensionality.

To optimize the network, we incorporate a batch normalization (BN) layer, dropout layer, and LeakyReLU layer following each deconvolutional layer. The BN layer standardizes the output of intermediate layers, thereby improving the model's training efficiency and effectiveness. The dropout layer mitigates overfitting within the generator, enhances the diversity and realism of the generated outputs, and addresses the gradient vanishing issue associated with traditional ReLU activation by implementing a LeakyReLU layer. This combination boosts the generator's robustness and generalization capacity. Furthermore, we introduce an attention layer immediately following the final deconvolutional layer to emphasize vital features. Finally, the dense layer applies the Tanh activation function to the output, ensuring optimal signal representation.

Attention mechanisms can enhance models' ability to focus on crucial features in input data, leading to improved accuracy and faster convergence. The self-attention mechanism is particularly effective at capturing intricate feature interactions by leveraging internal information. Specifically, the input feature $\mathbf{X}$ is first transformed into a triad ($\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$)

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using three trainable weight matrices: $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$. Subsequently, the similarity between $\mathbf{Q}$ and $\mathbf{K}$ is computed, and this similarity serves as a weighting coefficient for $\mathbf{V}$. Fig. 5 provides a visual representation of the self-attention mechanism's computational flow, while (12) and (13) outlines the attention mechanism's computation steps

$$\mathbf{Q} = \mathbf{W}^Q\mathbf{X} \tag{12.a}$$

$$\mathbf{K} = \mathbf{W}^K\mathbf{X} \tag{12.b}$$

$$\mathbf{V} = \mathbf{W}^V\mathbf{X} \tag{12.c}$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{13}$$

where d_k is the dimension of the feature vector. It is utilized as a divisor during normalization to prevent the calculation of an excessively large vector inner product.

We employ the LeakyReLU activation function in the generator network, and the hyperbolic Tanh activation function in the output layer. The LeakyReLU activation function is also used in the remaining layers. The LeakyReLU function is as follows:

$$\begin{aligned}\text{LeakyReLU}(x) &= \begin{cases} x, & x > 0 \\ \gamma x, & x \le 0 \end{cases} \\ &= \max(0, x) + \gamma \min(0, x). \end{aligned} \tag{14}$$

The Tanh function is a zero-mean function that improves the training efficiency, which is given as follows:

$$\text{Tanh}(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}. \tag{15}$$

The loss function of the generator is

$$V(G) = \mathbb{E}_{z\sim P_z(z)}[\log(1 - D(G(z)))] \tag{16}$$

where z is the RCG signal, with its prior probability given by $P_z(z)$. $G(z)$ is the discriminator's output for z, while $D[G(z)]$ represents the discriminator's output for the generated sample $G(z)$. This output can be interpreted as the likelihood that the generator's sample will be deemed genuine.

3) Network Structure of Discriminator: Fig. 6 depicts the proposed discriminator's network structure. The model's input is a reconstructed ECG-like signal fed through a dense layer. We employ four Conv1D layers following the dense layer. These layers perform feature extraction and transformations on the input using convolutional operations. After each Conv1D layer, we add a LeakyReLU activation function to introduce nonlinearity and enable the network to learn more complex data distributions. Finally, a dense layer is the output layer, activated using a sigmoid activation function that constrains the output to the range [0, 1]. The sigmoid function is defined as follows:

$$\text{sigmoid}(x) = \frac{1}{1 + \exp(-x)}. \tag{17}$$

The loss function of the discriminator is given by the following equation:

$$V(D) = \mathbb{E}_{x\sim P_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \tag{18}$$

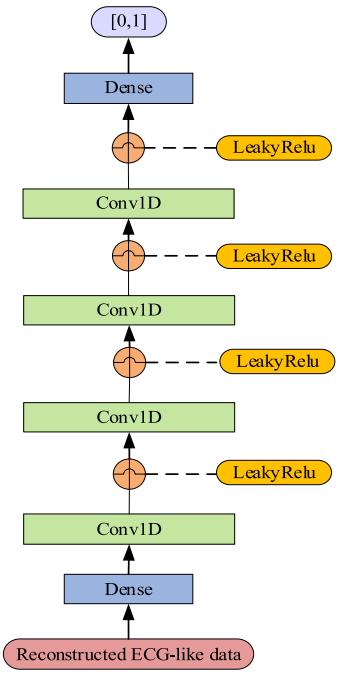


Fig. 6. Structure of the discriminator.

where $P_{\text{data}}(x)$ is the distribution of x, $D(x)$ denotes the output of the discriminator for the true sample x, and $D[G(z)]$ represents the discriminator's output for the generated sample $G(z)$ produced by the generator.

IV. Experimental Analysis and Discussion

A. Data and Platform

The dataset utilized in this study is the open dataset MMECG, provided by the University of Science and Technology of China [17]. This dataset encompasses heartbeat echo data and synchronized ECG data from ten participants across four distinct physiological states: normal respiration, irregular respiration, post-exercise (PE), and sleep (SP). Table I outlines the dataset structure, where the RCG and posXYZ represent the heartbeat signals and their corresponding heart locations. In addition, the dataset provides information on the participant's index (id), age, gender, and physiological status (physistatus). The physiological status categories include normal breathing (NB), irregular breathing (IB), PE, and SP. The Attention-DCGAN network is developed using the TensorFlow deep learning framework and trained on an Inter I7-10700K processor and an NVIDIA GTX3090 graphics card configuration.

B. Data Preprocessing

To enhance the signal-to-noise ratio of radar signals, we employ an improved K-means clustering algorithm to categorize 50 3-D heart locations' radar signals. This process generates the RCG signals from the simultaneous ECG measured by the millimeter-wave radar. The clustering outcomes are displayed in Fig. 7. As observed in Fig. 7, by implementing the enhanced K-means clustering algorithm, we further integrate the signals into the cluster

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TABLE I
DATASET PARAMETER CONFIGURATION

parameter	value
RCG	35505×50 double
ECG	35505×1 double
posXYZ	50×3 double
id	1、2、5、9、13、14、16、17、29、30
gender	girl/boy
age	22、23、24、25、29、37、45、65
physistatus	NB/IB/PE/SP

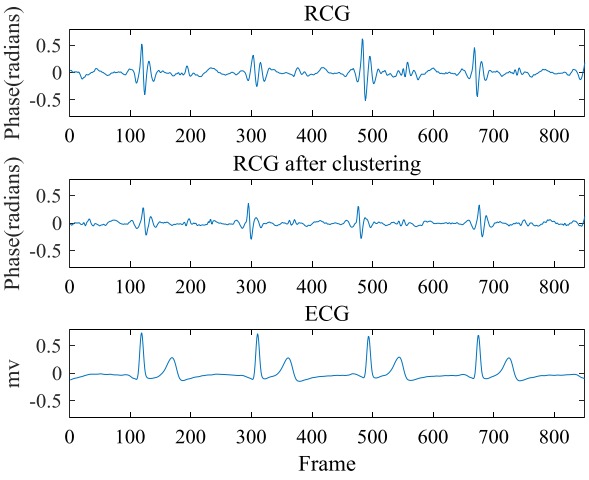


Fig. 7. RCG signals after clustering.

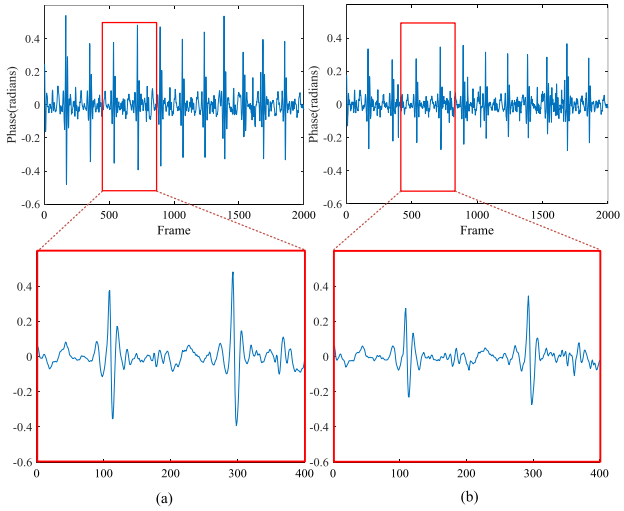


Fig. 8. Comparison before and after clustering. (a) Before clustering. (b) After clustering.

centers, which elevates the quality of the radar measurement data. Before clustering, numerous neighboring signals are grouped in the time domain with substantial fluctuations, as depicted in Fig. 8(a). Following clustering, the signals [illustrated in Fig. 8(b)] exhibit greater discriminability and stability in the time domain representation, enhancing the accuracy of the millimeter-wave radar's heartbeat information assessment.

TABLE II
CONVOLUTIONAL LAYER MODULE CONFIGURATION OF DISCRIMINATOR

Type	$k\times$kernel size	Stride	Output size (Batch, Channel, Length)
Input	-	-	(32, 1, 200)
Conv1d-1	512×5	1	(32, 1, 512)
Conv1d-2	256×5	1	(32, 1, 256)
Conv1d-3	128×5	1	(32, 1, 128)
Conv1d-4	64×5	1	(32, 1, 64)

TABLE III
CONVOLUTIONAL LAYER MODULE CONFIGURATION OF GENERATOR

Type	$k\times$kernel size	Stride	Output size (Batch, Channel, Length)
Input	-	-	(32, 1, 200)
Conv1d_transpose-1	512×5	1	(32, 1, 512)
Conv1d_transpose-2	256×5	1	(32, 1, 256)
Conv1d_transpose-3	128×5	1	(32, 1, 128)
Conv1d_transpose-4	64×5	1	(32, 1, 64)

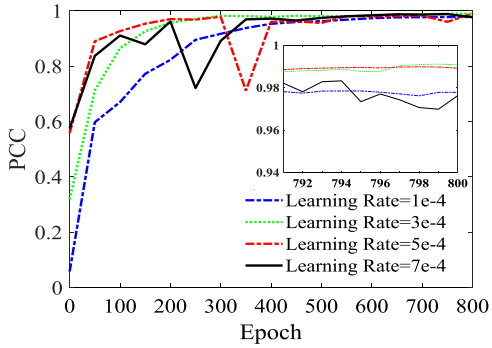


Fig. 9. PCC curves for Attention-DCGAN networks with different initial learning rates.

C. Hyper-Parameters Selection

After applying K-means clustering to the RCG signal and partitioning the original ECG signal into segments centered on the R peaks with a length of 200 samples, we can obtain signal fragments corresponding to complete heartbeat cycles. These fragments serve as inputs to our GAN model: the generator receives the isolated heartbeat signal fragments, while the discriminator is fed with the synchronized ECG signal fragments.

Tables II and III present the initialization configurations for the convolution modules used in the generator and discriminator, where k indicates the number of convolution kernels. During training, we utilize a batch size of 32 over 800 iterations, sourced from a dataset containing 2000 samples. To mitigate the risk of converging to local minima, we investigate various initial learning rates during both the training and testing phases. As shown in Fig. 9, the Pearson correlation coefficient (PCC) analysis indicates that the most accurate reconstruction is achieved with an initial learning rate of 0.0003, which we subsequently adopted for further experiments.

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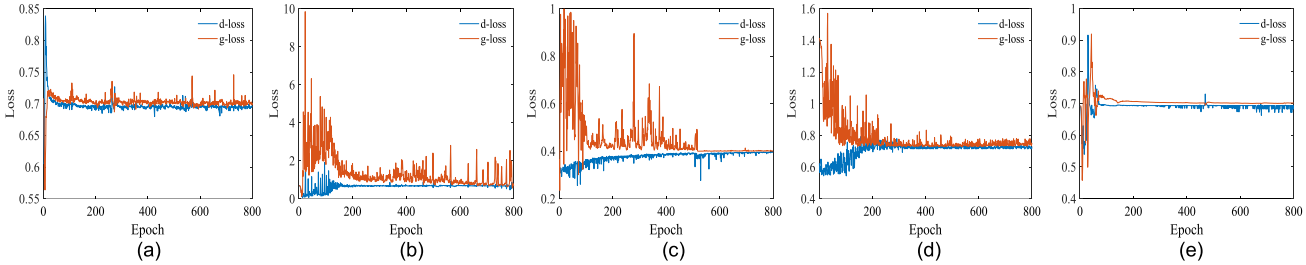


Fig. 10. Loss function of five networks. (a) GAN. (b) DCGAN. (c) AdvGAN. (d) SpeechAdvGAN. (e) Attention-DCGAN.

D. Performance Comparison

We have evaluated four types of GAN architectures: two baseline models—GANs and DCGANs—alongside two contemporary architectures: adversarial examples with GANs (AdvGAN) [28] and Speech with GANs (SpeechAdvGAN) [29]. While retaining the foundational architecture of AdvGAN and SpeechAdvGAN, we have modified certain parameters to align with our data format.

The architectures of the two baseline models are as follows.

Baseline Network I (GAN): The generator comprises four fully connected layers. The activation functions for the hidden layers are LeakyReLU, except for the output layer, which uses the Tanh function. The discriminator consists of four fully connected layers, with LeakyReLU activation functions in the hidden layers and a sigmoid function in the output layer. To prevent overfitting, a Dropout layer is added to the discriminator.

Baseline Network II (DCGAN): The generator is a convolutional neural network with six layers. It starts with a fully connected layer, followed by four anti-convolutional layers that gradually increase the data dimension of the output. Finally, an additional fully connected layer produces the output. The Dropout layers are inserted after each anti-convolutional layer, LeakyReLU is used as the activation function, and BN is applied. The discriminator also has a six-layer convolutional neural network structure, with a fully connected layer at the input and four convolutional layers following it. LeakyReLU is used as the activation function in these layers, and a final fully connected layer activates the output with a sigmoid function.

E. Training Results of the Networks

The training dynamics of the generator and discriminator are depicted in Fig. 10, where the d-loss curve corresponds to the discriminator's loss, and the g-loss curve corresponds to the generator's loss. Fig. 10(a) and (d) illustrate that the training processes of GAN and SpeechAdvGAN become increasingly stable over time, albeit with some residual fluctuations. Similarly, Fig. 10(c) shows that AdvGAN, which initially experiences large fluctuations, stabilizes as training progresses. In contrast, the training process of DCGAN, shown in Fig. 10(b), remains unstable. Our experiments reveal that incorporating a BN layer can accelerate the training of the DCGAN network, but it also introduces instability in the generator's training. To address this, we opted not to apply BN to the final layer of the Attention-DCGAN generator.

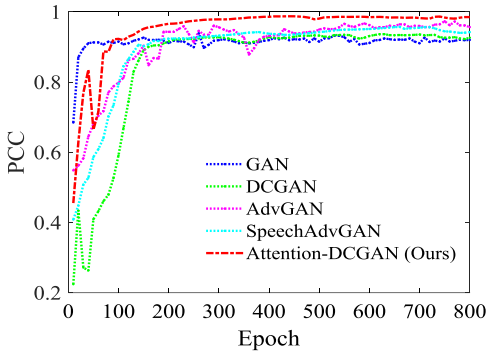


Fig. 11. PCC curves of five models.

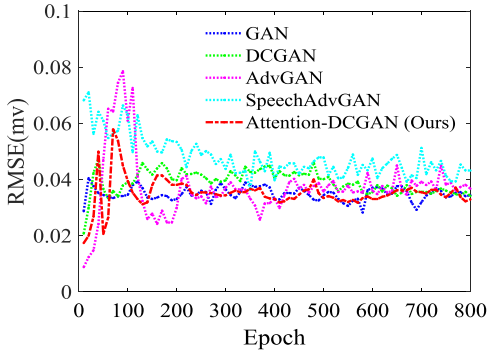


Fig. 12. RMSE curves of five models.

This adjustment enhances training efficiency while preserving stability. Following these improvements, we observe a steady decrease in the generator's loss and stabilization of the discriminator's loss, signifying enhanced model performance.

We evaluate the PCC and RMSE between the generated data and the original ECG data, as presented in Figs. 11 and 12, respectively. Fig. 11 demonstrates that the PCC of Attention-DCGAN increases consistently with iterations, nearing a value of 0.9853. In contrast, GAN, DCGAN, AdvGAN, and SpeechAdvGAN exhibit lower correlation coefficients, with AdvGAN showing significant fluctuations. As shown in Fig. 12, the RMSE of Attention-DCGAN remains stable and low, around 0.0329 mV, as training batches increase.

Fig. 13 compares the original RCG data, original ECG data, and reconstructed ECG-like data of the Attention-DCGAN. It indicates that the improved generator produces data closely resembling the original ECG waveform. Consequently, Attention-DCGAN efficiently reconstructs ECG-like signals from RCG signals with high accuracy.

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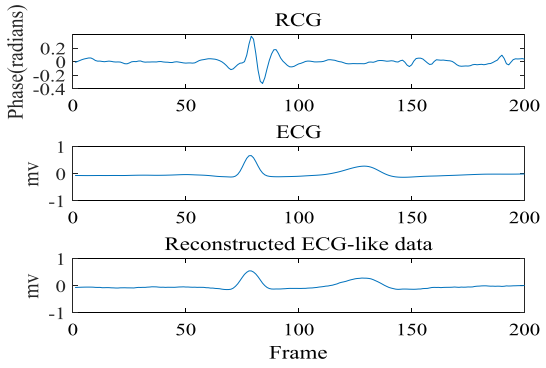


Fig. 13. Comparison of cardiac signal reconstruction.

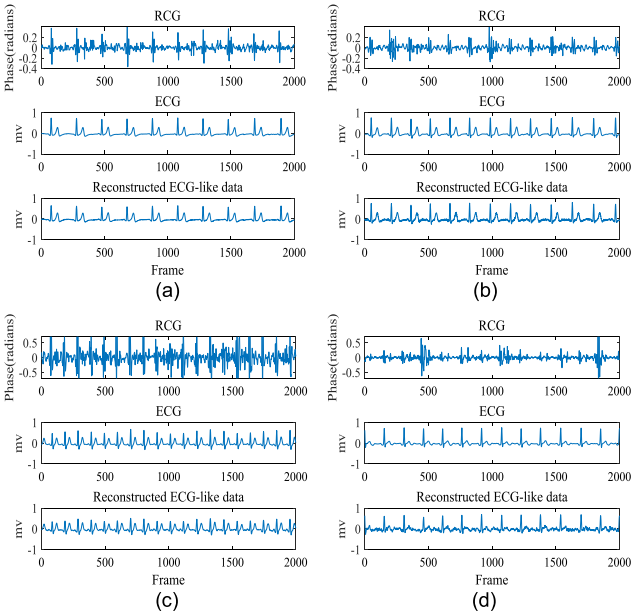


Fig. 14. RCG, ECG, and reconstructed ECG-like data of four states. (a) "NB." (b) "IB." (c) "PE." (d) "SP."

TABLE IV
COMPARISON OF ECG SIGNAL RECONSTRUCTION RESULTS UNDER DIFFERENT METHODS

Method	PCC	RMSE(mV)
GAN [25]	0.9202	0.0344
DCGAN [26]	0.9265	0.0342
AdvGAN [28]	0.9568	0.0368
SpeechAdvGAN [29]	0.9425	0.0432
Attention-DCGAN (ours)	**0.9853**	**0.0329**

F. Test Results of the Networks

We employ 2000 frames of ECG signals from the MMECG dataset, with the results in Fig. 14. This figure illustrates the RCG data, original ECG data, and reconstructed ECG-like data corresponding to the four states, highlighting the method's strong generalization performance. To further validate our approach, we applied GAN, DCGAN, AdvGAN, SpeechAdvGAN, and Attention-DCGAN to signal reconstruction across 100 test samples, calculating their correlation coefficients with the real ECG signals, as detailed in Table IV. The table clearly shows that Attention-DCGAN delivers superior reconstruction accuracy for ECG-like signal.

V. CONCLUSION

This article presents a novel approach for minimizing the disparity between millimeter-wave radar heartbeat signals and ECG signals by reconstructing RCG signals into ECG-like waveform based on the public MMECG dataset. The methodology consists of two primary steps. First, to address fluctuations in the time-domain waveform, we employ an improved K-means algorithm to cluster the RCG signals. This clustering enhances the data's identifiability and stability, leading to more accurate signal processing. Second, we develop a DCGAN integrated with an attention mechanism specifically designed for RCG signals. This network leverages the dynamic interaction between the generator and discriminator to iteratively refine the generated RCG waveforms, making them closely resemble real ECG signals. The incorporation of a self-attention mechanism enables the network to effectively capture correlations across different positions within the long sequence of periodic RCG signals, significantly improving reconstruction accuracy.

Experimental results demonstrate that the proposed Attention-DCGAN achieves stable and precise reconstruction of RCG signals into ECG-like signals, with an RMSE of 0.0329 mV and a PCC of 0.9853. These findings validate the feasibility of reconstructing ECG-like signals from millimeter-wave radar signals, offering a more accurate solution for radar-based ECG measurement applications. Furthermore, the method's adaptability to other datasets highlights its potential value in medical diagnosis.

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